

DEUTSCHE BAHN

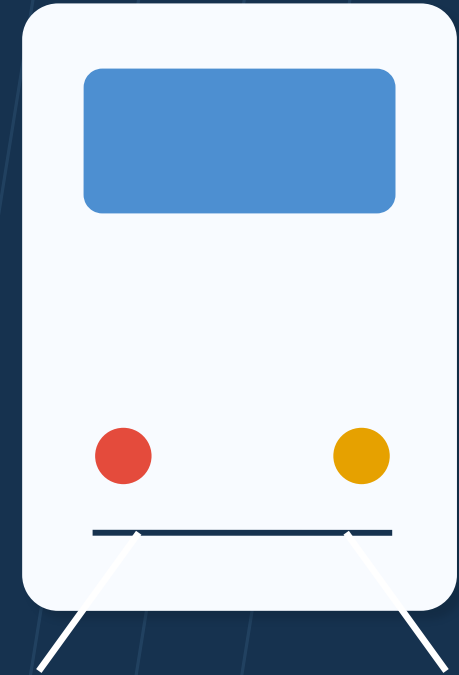
Train Delay Analysis

Final Data Visualization Project

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Matriculation No. 82022509
Subject: DATA VISUALIZATION

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Project at a glance

A compact summary of the cleaned dataset, analytical focus and overall reliability picture.

76,476

cleaned service records

20

major German stations

8

sampled dates

44.4%

overall delay rate

3.0 min

median actual delay

Core research question

How do train delays differ by station, train category, time, sampled date and recorded disruption reason across major German railway stations?

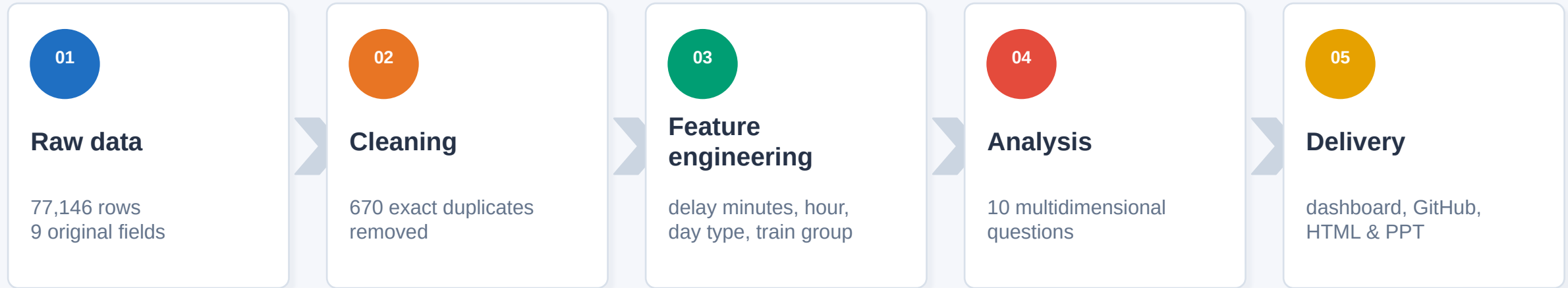
The analysis separates delay frequency from delay severity, so high-volume stations are not automatically treated as the least reliable.

What the project delivers

- 10 analytical questions
- 10 explanatory Plotly visualisations
- Interactive Streamlit dashboard
- Public GitHub repository
- Notebook export and presentation

From raw train records to decision-ready insights

The workflow preserves the original data, creates interpretable features and then tests multiple reliability dimensions.



Key engineered variables

`actual_delay_minutes`

`train_group`

`hour`

`day_type`

`platform_change`

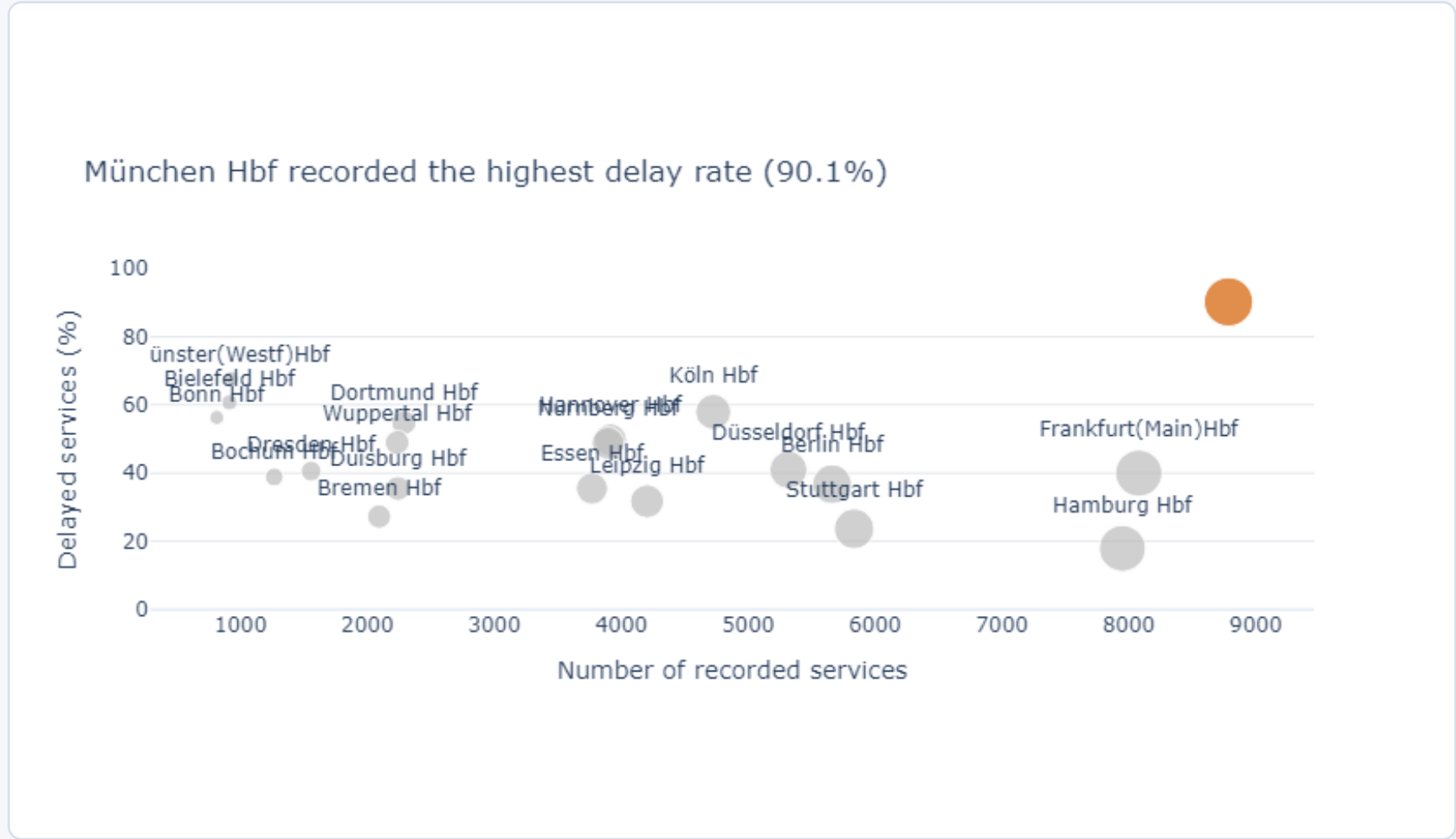
`delay_reason`

The original CSV remains unchanged in data/raw; the cleaned file is saved separately in data/processed.

FINDING 1

Station reliability varies far more than service volume alone can explain

München Hbf combines one of the highest service volumes with the highest recorded delay rate.



Key takeaways

90.1%

München Hbf delay rate

67.6%

Münster (Westf) Hbf

17.9%

Hamburg Hbf despite high volume

Interpretation

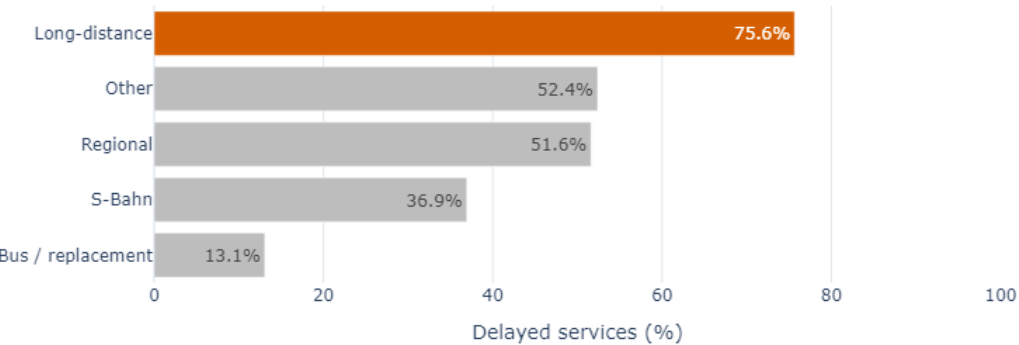
Delay rate gives a fairer comparison than delayed-service counts alone.

FINDING 2

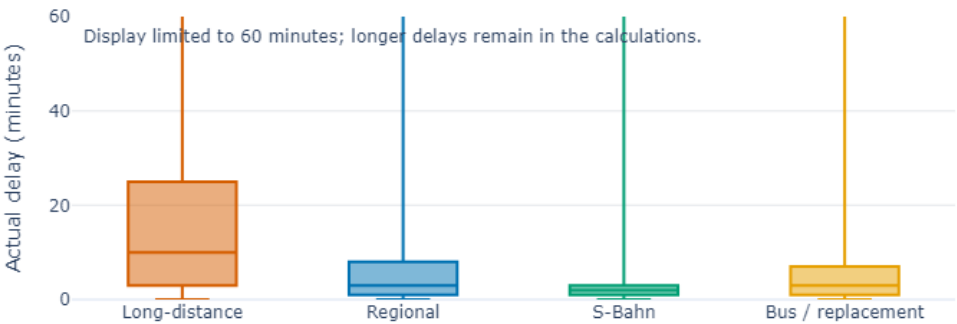
Long-distance services are the core reliability issue

They are delayed most often and, when delayed, typically remain late for longer.

Long-distance services had the highest delay rate (75.6%)



Long-distance delays were substantially more severe than other service types



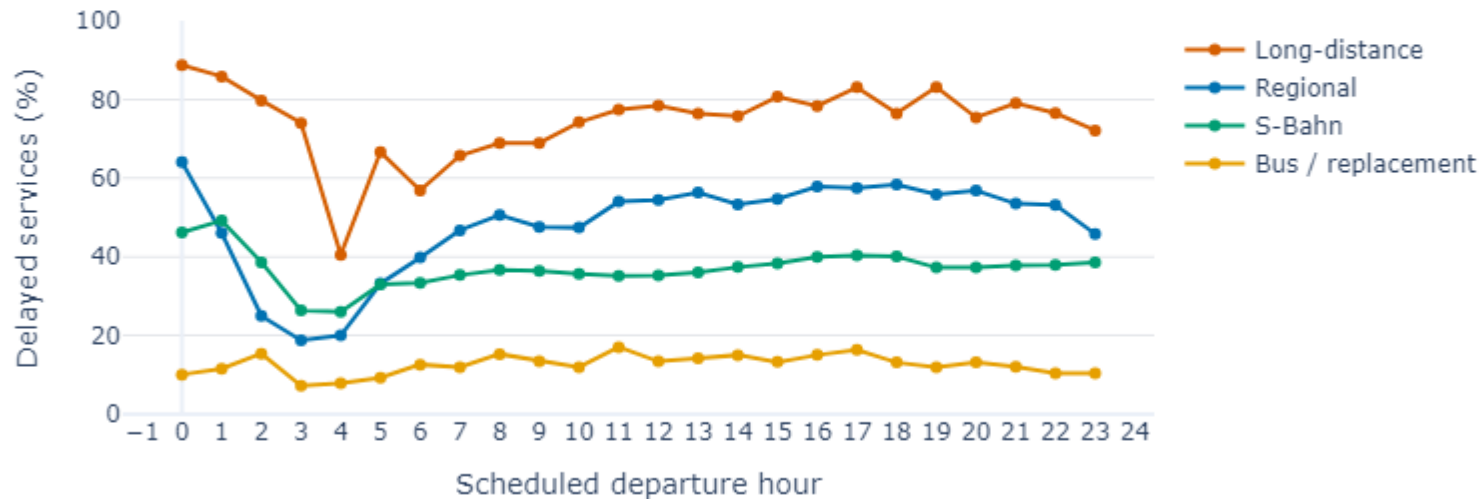
Frequency: 75.6% of long-distance records were delayed. **Severity:** median actual delay was about 10 minutes, versus 2-3 minutes for the other main groups.

FINDING 3

Long-distance services remain the most delay-prone throughout the day

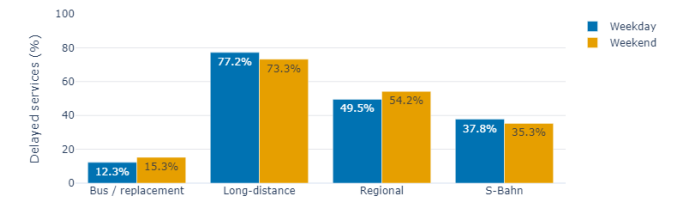
The gap between service types is more persistent than any single peak-hour effect.

Long-distance services were the most delay-prone at every hour



Weekday vs weekend

Weekend effects differed by service type rather than moving all categories together



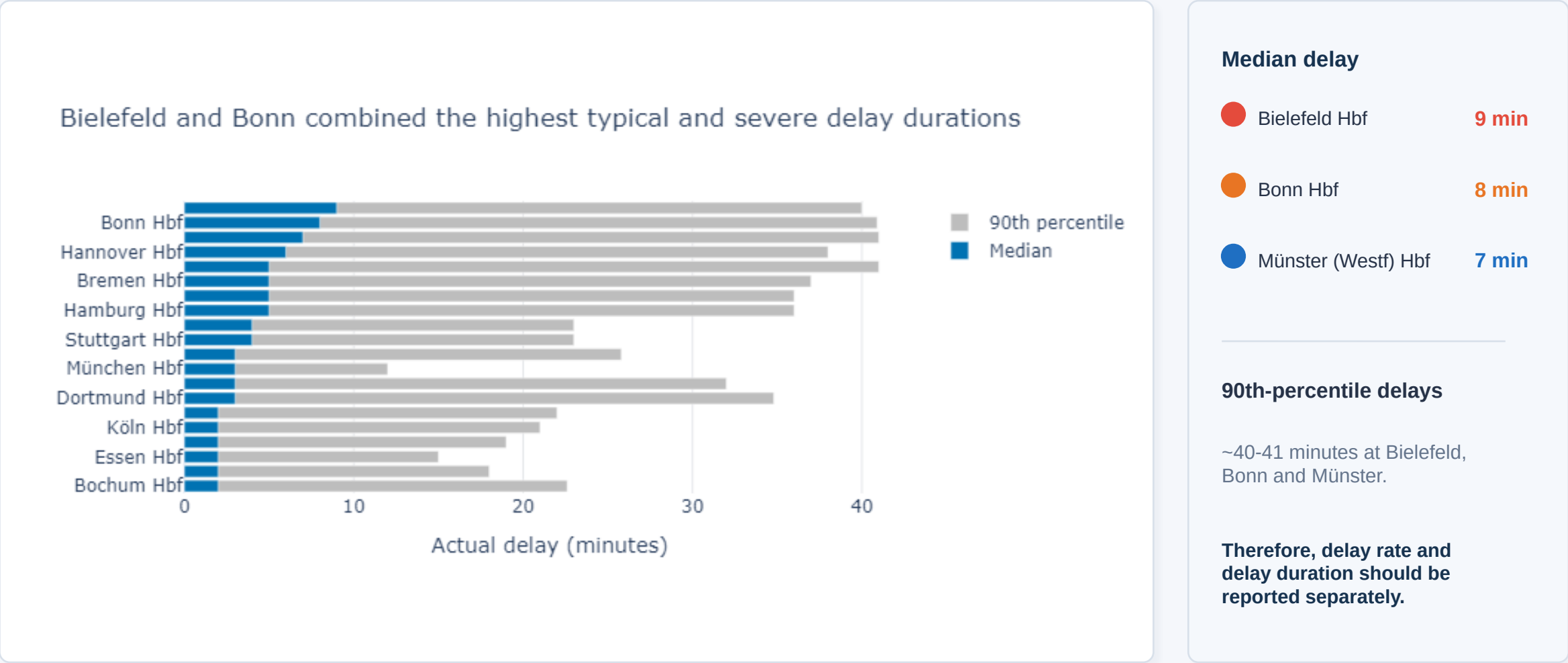
What changes?

- Long-distance: slightly worse on weekdays
- Regional: higher delay rate on weekends
- S-Bahn: small weekend improvement

FINDING 4

The “worst” station changes when delay severity is considered

München leads on delay rate, but Bielefeld and Bonn show longer typical delays.

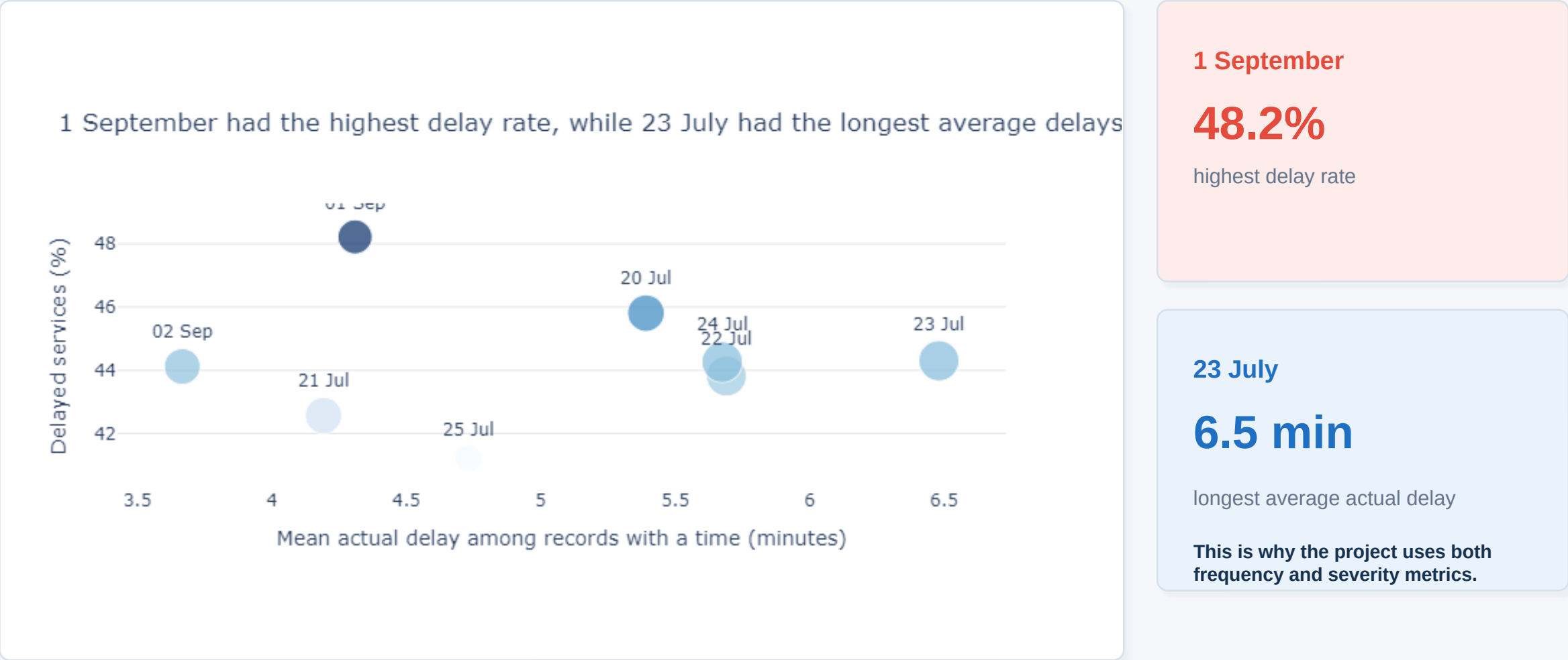


Question 7 - median and 90th-percentile actual delay by station

FINDING 5

Delay frequency and delay duration peak on different days

No single date was worst on every reliability measure.



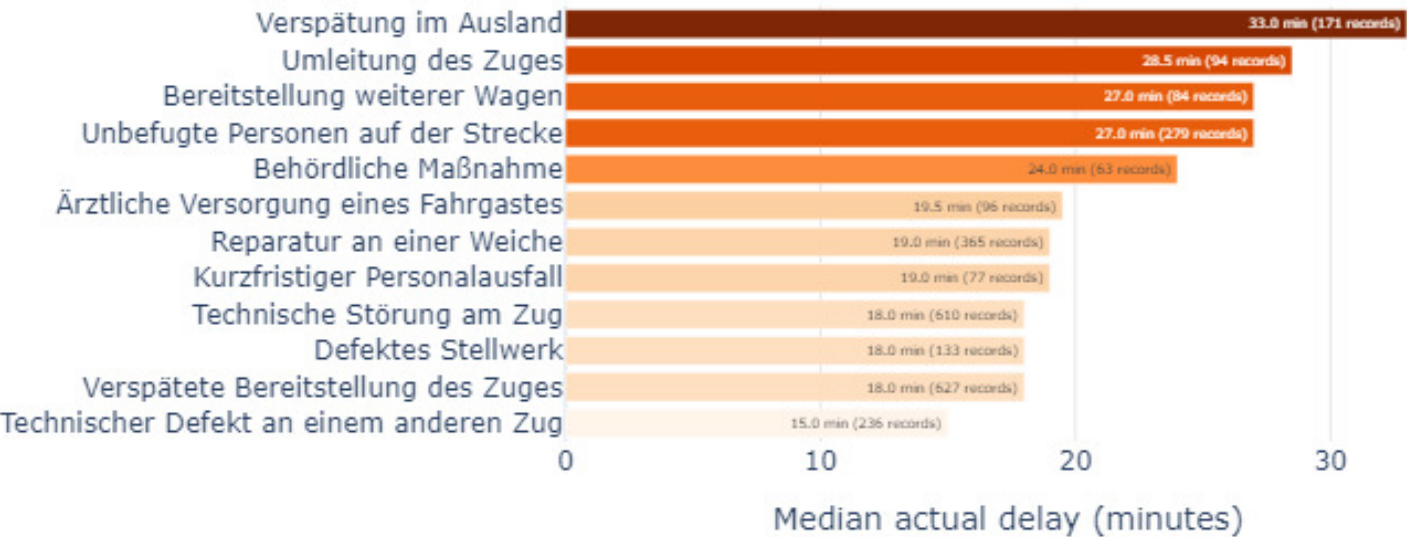
Question 8 - sampled dates compared by delay rate, average delay and service volume

FINDING 6

Less-common disruptions can produce the most severe delays

The most frequent operational issue is not necessarily the most damaging.

International lateness and train diversions were linked to the longest typical delays



Highest median delays

33.0 min

Delay originating abroad

28.5 min

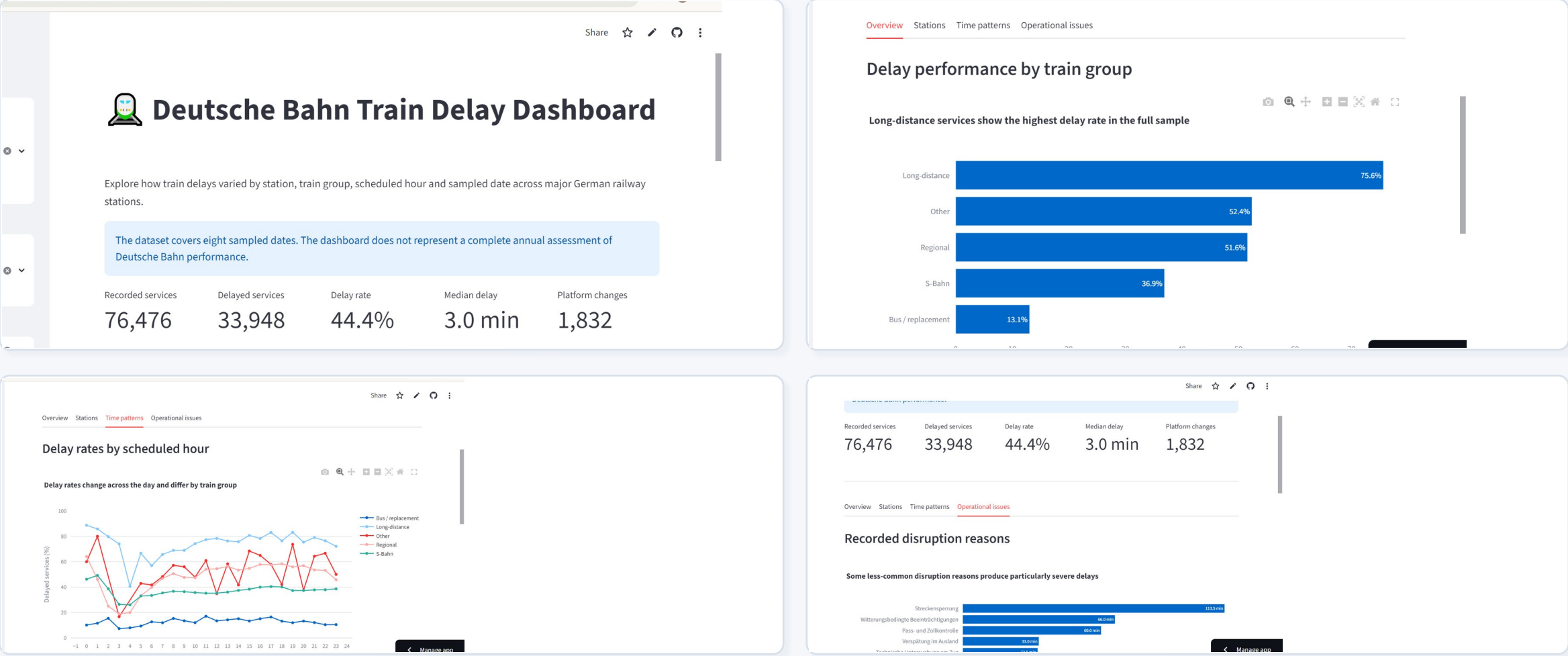
Train diversion

Platform changes

Münster recorded the highest platform-change rate (10.9%), while München changes carried a median delay of about 11 minutes.

Interactive dashboard: from static findings to guided exploration

Users can filter by station, train group, day type, sampled date and scheduled hour.



Live dashboard: deutsche-bahn-delay-analysis-7rj2tgss6za6aykj3zjnr.streamlit.app

Conclusions, limitations and project access

The analysis is strongest when frequency, severity, station context and operational causes are considered together.

Main conclusions

- Long-distance services were the weakest-performing category.
- München had the highest delay rate, while Bielefeld and Bonn had longer typical delays.
- Delay frequency and delay severity identify different problem areas.
- Operational causes have unequal effects; international lateness and train diversions were especially severe.

Key limitation

The dataset covers only eight sampled dates, so the findings should not be treated as a full annual assessment of Deutsche Bahn performance.

Open the project



[GitHub repository](#)



[Live dashboard](#)

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Thank you